

Sex related differences in Medicare care and health IT opportunities

Evidence synthesis and methods for a systematic review in progress

September 21 2026

Abstract

Objectives: To examine sex-related differences in Medicare coverage, service valuation, patient costs and access, and identify questions that researchers and health IT developers can test.

Methods: This synthesis combines a selected register with PubMed development searches and a separate policy and implementation assessment. Searches identified 59,147 unique records after exact-PMID deduplication. Independent human screening, full eligibility adjudication and formal appraisal remain incomplete. Patient sex, physician gender, clinical anatomy, payment and utilization were analyzed as distinct concepts.

Results: The working register contains 38 candidate or contextual records, not 38 systematically included studies. Selected Medicare studies associate female sex with cost-related nonadherence, but estimates vary by outcome and population. Procedure-valuation findings depend on matching and payment measures. Bone-care studies identify delivery gaps affecting women and men. Intervention evidence supports testing some outreach and affordability approaches, while null benefit-tool findings challenge assumptions that information alone improves access. Six opportunity domains are proposed without profitability rankings.

Conclusions: Health IT opportunities should be selected by demonstrable patient need, intervention benefit, equitable reach and feasible implementation. Coverage gaps or payment differences alone do not establish bias, product effectiveness or commercial viability. This manuscript requires completion of the systematic-review procedures before submission as a completed review.

Key takeaways

- Medicare coverage, clinician payment, patient cost and receipt of appropriate care are different outcomes; a product must identify which mechanism it can change.
- Evidence about an intervention's average benefit does not establish that it reduces a sex-related inequity. Equity effects require explicit comparisons and adequate measurement.
- Medication affordability, bone-care coordination, authorization workflows, valuation research, lymphedema fulfillment and kidney-care coordination warrant different kinds of validation before investment or deployment.

Keywords: Medicare; health equity; sex differences; reimbursement; health information technology; evidence synthesis.

Introduction

Differences in health care for women and men can arise from clinical need, reproductive anatomy or physiology, social circumstances, professional practice, and financing. The Lancet review by Mauvais-Jarvis and colleagues provides an organizing account of these biological and social pathways, but it is not an analysis of Medicare reimbursement. Its framework supports a broad inquiry beyond reproductive medicine, including cardiovascular, renal, respiratory, metabolic, musculoskeletal and mental health. Each underlying empirical claim requires separate assessment. [L-B01; L-B01-E1]

Medicare offers a national financing benchmark for care in later life and for younger people with qualifying disabilities or conditions. It does not finance most US health care, and spending is not a measure of clinical need. Cross-sectional age-specific expenditure data cannot establish a lifetime share of care. This review therefore distinguishes utilization, financial expenditure and unmet need, while retaining younger beneficiaries and entitlement categories. Original Medicare, Medicare Advantage and Part D can expose otherwise similar patients to different benefit arrangements and delivery systems. [P-041; P-042; P-043]

For researchers and developers, the practical question is not simply whether reimbursement differs. It is whether an identifiable financing or delivery mechanism prevents beneficial care, whether an intervention can change that mechanism, and whether improvement reaches patients equitably. A software company can streamline a transaction without improving health; an intervention can improve average access while widening disparities. Conversely, a clinically worthwhile service need not have a higher-paying counterpart for another sex.

This manuscript develops an evidence-based research agenda from the available register and newly retrieved implementation studies. It makes the remaining systematic-review work explicit. It neither estimates the national prevalence of reimbursement bias nor identifies a proven commercial market. The opportunity assessment is an interpretation of selected evidence, developed after that evidence was inspected.

Methods

Design and review questions

The intended study is a systematic review of empirical Medicare coverage, service valuation, patient spending and access outcomes, with a separately identified translational assessment. The current report is a preliminary evidence synthesis from a systematic review in progress. PRISMA 2020, PRISMA-S and PRISMA-Equity guide the planned reporting; PRESS informs independent search review. These are reporting and methodological frameworks, not certificates of study quality or acceptance by a journal. [M-001; M-004; M-005; M-006]

Three questions organize the empirical review. First, among patients with comparable clinical need, how do coverage, costs and receipt of appropriate care differ by recorded sex or gender? Second, how are sex-associated services, including shared-system and reproductive care, valued relative to defensible clinical and resource comparators? Third, what evidence links a financing or administrative mechanism to a care gap? A fourth, explicitly separate question asks which interventions might address an established gap and what validation would be required. Physician-gender payment studies provide context and are not pooled with patient comparisons. Figure 1 separates the analytical subjects and their inference boundaries.

Population and clinical classification

The investigator's existing scope concerns cisgender women and men and excludes transgender-focused patient studies and gender-affirming-care indications. Administrative male/female coding does not establish gender identity. Under the current rule, studies with unreported identity remain provisional candidates, outside confirmed-cisgender primary estimates. No individual is classified through anatomy, diagnosis or medicine use.

Mixed-payer or mixed-population studies require separately extractable eligible results; their presence alone does not establish Medicare-specific effectiveness. Figure 2 distinguishes adopted population scope, proposed screening rules and unresolved eligibility.

Care is classified as shared systems or functions, female reproductive anatomy, male reproductive anatomy, female reproductive physiology, male reproductive physiology, or mixed/site-dependent services. Bone, breast, lymphatic and renal care can concern shared systems even when disease burden or eligibility differs by sex. Menopause is a physiological context; a hormone name alone does not establish the anatomical category. Sex-defined policy eligibility is recorded separately from anatomy and clinical need. Bundled or site-dependent procedures require component-level review. Figure 3 maps these clinical categories and the separate policy and measurement axes.

Clinical need is defined independently of reimbursement rules. Otherwise, people denied a benefit could disappear from the denominator used to assess inequity. Lower utilization is not automatically undertreatment; higher utilization is not automatically better care. Appropriate comparisons require indication, contraindications, patient preferences, alternatives, follow-up and outcomes.

Searches and record management

PubMed development searches were executed through NCBI E-utilities on September 21, 2026, before independent PRESS review. The corrected Medicare/payment/access core returned 43,904 records and the supplementary sex/gender valuation/cost query returned 16,866. Removal of 2,084 overlapping PMID hits produced 58,686 unique records. The initial core used an unrecognized formulary MeSH heading; correction to Formularies as Topic added 60 records and removed none. Original queries and exports were preserved.

Two later, targeted development extensions addressed a missed bone-testing source and the new health IT purpose. They returned 494 bone-care records and 262 prescription-information records. Of these 756 records, 295 overlapped the baseline and 461 were additional, yielding 59,147 unique PMIDs. The previously missed bone-testing paper was retrieved by the extension. This diagnostic repair does not establish comprehensive recall. No language, age, design or full-text availability filters were imposed. Technical date partitions reconciled ESearch output limits against unrestricted counts.

Raw responses, submitted queries, translations, warnings, timestamps, checksums and PMID memberships are archived. All combined records have bibliographic metadata; 47,152 contain abstracts and 11,995 do not. Exact-PMID deduplication is complete for these queries. Computational checks flagged 17 shared-DOI groups and 303 normalized-title groups for inspection; these potentially overlapping categories were not automatically merged. Distinct reports, corrections and overlapping cohorts must be adjudicated separately.

A transparent keyword queue orders records for review and retains every record. It is not a validated classifier or an exclusion rule. The prior 50-record calibration set remains available with blank human decision fields. The 38-record working register was assembled through earlier selected retrieval and cannot be treated as the output of screening the 59,147-record corpus. Targeted intervention retrieval and policy searches are separate source routes requiring reconciliation. Searches of the other planned databases and independent PRESS review remain pending. Figure 4 shows the executed retrieval state without implying completed selection.

Selection extraction and appraisal

Two human reviewers are planned to independently screen records, reconcile disagreements and document reasons for full-text exclusion. They will independently extract critical outcome data and make design-specific appraisal judgments before reconciliation. No such completed dual review is claimed. Original studies, companion reports, background reviews, clinical efficacy evidence and physician-gender context will retain distinct roles. Citations from a review will not become included studies merely because they were cited.

Scientific metadata are verified separately from access, eligibility and appraisal, with numeric PMIDs retained whenever available.

Extraction records include study/report identifiers, design, payer, entitlement, period, population, sex/gender measurement, clinical need, exposure, comparator, outcome, denominator, missingness, effect estimate, uncertainty, adjustment variables, funding and source location. Full-text availability varies and is disclosed. Newly reviewed interventions are assessed as implementation context unless they meet the primary Medicare-equity criteria. Reports sharing a trial are linked to prevent double counting.

Planned appraisal uses design- and result-appropriate instruments, including RoB 2 for randomized intervention effects and the revised JBI instrument for analytical cross-sectional associations. Exact versions and interpretation rules will be documented and piloted before appraisal. Observational disparities require attention to selection, clinical-need measurement, confounding and possible mediation; other designs require appropriate instruments. Policy and code comparisons require checks of authority, currency, matching, resource definitions and reproducibility. Provisional AI extraction notes do not constitute independent human appraisal or a numeric quality score. [M-010; M-011]

Synthesis and opportunity assessment

Coverage, allowed payments, work RVUs, total RVUs, physician receipts, patient spending and realized access are synthesized separately. Historical observations are aligned to their policy period. Figure 5 distinguishes financing measures, care access and health outcomes; connections are hypotheses requiring evidence. Within-condition patient contrasts are separated from cross-procedure comparisons and assessments within a sex-linked indication. Null, contradictory and imprecise findings receive the same interpretive attention as favorable results. No meta-analysis is performed because compatible, independently adjudicated outcome groups have not been established. SWiM will inform suitable non-meta-analytic intervention syntheses; GRADE or CERQual will be applied only to appropriate outcome-specific bodies of evidence, not to individual papers as a score. [M-002; M-007; M-008]

The translational assessment separates evidence of a problem, evidence that an intervention helps, and evidence that implementation or purchasing is feasible. For each domain it records the intended patient, clinical mechanism, workflow, prospective user and buyer, data needs, policy status, alternatives, measurable benefit, harms and validation design. Proposed buyers and business models are hypotheses. No market-size estimate, revenue forecast, composite opportunity score or profitability rank is asserted. Findings about a mandated API do not establish payment for software. These assessment rules were developed after the exploratory register and the investigator's entrepreneurship request; they are not prospectively registered analyses.

Results

Evidence available and its boundaries

The active register comprises 38 candidate or contextual records spanning patient care, service valuation, physician gender and broad clinical frameworks. Clinical efficacy or other-payer studies within that register support background reasoning; they do not become eligible Medicare-equity studies through inclusion in the register. Eight earlier scope-excluded records and the Lancet correction remain in the audit trail. No final included-study count or completed PRISMA selection flow is available. Table 1 summarizes selected examples and their inference limits. Figure 4 presents the actual retrieval state and keeps selected evidence separate from the unscreened corpus.

Medication affordability and formulation specific access

In the 2016 Medicare Current Beneficiary Survey, the adjusted female-to-male odds ratio for any cost-related medication nonadherence was 1.66 (95% CI, 1.31-2.10) below age 65 and 1.21 (1.04-1.41) at age 65 or older. Reported prevalence was 40.5% versus 28.5% in the younger stratum and 15.7% versus 12.8% in the older stratum. A selected Chicago cohort instead produced an imprecise adjusted persistent-nonadherence odds ratio of 1.36 (0.95-1.95), while its transient-nonadherence estimate was 0.81 (0.62-1.07). Annual occurrence and longitudinal persistence are different outcomes. Neither study establishes current coverage discrimination or a causal software solution. [L-R01; L-R02]

Vaginal-estrogen prescribing data demonstrate Medicare Part D use and spending for selected products, contradicting a blanket claim that Medicare never reimburses estradiol. Spending per treated beneficiary is not the patient's out-of-pocket burden. Clinical studies also caution against assuming interchangeable effects across hormonal formulations, outcomes and populations. A trial of a particular vaginal estradiol regimen found no greater improvement in its principal symptom outcome than dual placebo, while a broader review found possible benefits for selected symptoms with generally limited certainty. These clinical results require proposed decision support to preserve indication and formulation differences; they do not establish the effectiveness of software. [L-D01; L-D03; L-D04]

A separate exploratory CMS formulary analysis in this project found at least one matched generic oral estradiol and vaginal-cream option in all 5,499 analyzed public plan segments in the August 2026 release. This is an original descriptive companion analysis, not a pooled review result. Its denominator is plan segments, not enrollment, and a listing does not establish suitability, affordability or a paid claim. Erectile-dysfunction-use exclusions are indication-specific; sildenafil used for another covered indication is not evidence of universal Viagra coverage. The original motivating comparison therefore needs replacement with precise questions about formulation, indication, benefit rules and patient costs. [P-002; P-003; P-026; P-034; P-037; P-038]

Service valuation depends on the comparison

A study of 55 selected procedure pairs reported higher work RVUs for male-associated services on average, but another comparison reported no statistically significant aggregate difference in work RVUs or RVUs per minute. The latter's journal abstract and companion summary disagree on pair counts, and that discrepancy remains unresolved. A time-based analysis found higher median work RVUs per hour for female-specific procedure labels but lower modeled dollars per hour when specialty compensation multipliers were applied. The multipliers were not Medicare conversion factors. [L-P01; L-P02; L-P03]

These findings are not interchangeable estimates of a single national effect. Pair selection, global periods, postoperative work, valuation year, surgeon work, setting and specialty compensation can change the estimand. A null comparison does not establish equivalence. A difference in RVUs does not establish patient denial, and clinician payment totals do not establish unequal rates for identical work. Procedure-valuation research is therefore an opportunity to improve comparability and measurement before making claims about correction of bias.

Bone care and shared systems

Historical Medicare evidence reports substantially lower DXA use in men than women. In a newer Asian American Medicare subgroup, crude DXA receipt was 19.8% for women and 5.0% for men, but study-entry age cutoffs differed: women were at least 65 and men at least 70. The adjusted result described in the abstract concerned race, not a sex effect. A separate women-only Medicare Advantage cohort reported two-year bone-measurement rates of 26.8% at ages 65-79 and 12.9% at age 80 or older. It cannot estimate a male-female disparity. These findings concern shared bone physiology, differing need and care delivery; they do not prove that Medicare coverage caused the gaps. [L-G04; L-R03; L-R04]

An outreach evaluation among men with a fragility fracture reported BMD testing of 10.7% versus 4.9% and osteoporosis-treatment initiation of 4.0% versus 2.5%. The analysis retained 7,842 of 10,934 randomized participants, excluding 28.3% after assignment. This weakens inference from randomization. The intervention combined patient and physician communication rather than isolating a software component. Improved testing or initiation is a process benefit; the study does not establish a reduction in fractures or a sex-gap-closing effect. It offers a concrete pathway for a prospective coordination trial, with eligibility, completion and downstream clinical outcomes measured separately. [L-G05]

Physician gender and other contextual streams

Studies of ophthalmologists, cardiologists and surgeons describe differences in annual Medicare receipts by physician gender. Their units are clinicians or services, not patient-sex access. The surgeon study also found an unadjusted colorectal payment-per-service comparison that was not statistically significant. These results belong in a distinct workforce and practice-pattern stream and should not be used to claim that women patients receive less covered care. [L-C01; L-C02; L-C03]

Lymphedema cost and insurer-policy studies, and qualitative kidney-care interviews, identify additional questions about fulfillment and coordination. Their Medicare directness and policy timing are limited. Historical lymphedema findings predate the 2024 Medicare compression-treatment-item benefit. Kidney-care interviews describe clinician perceptions rather than measured patient or Medicare outcomes. These domains require contemporary needs assessment before claims about product benefit. [L-A01; L-A02; L-Q01]

Evidence that proposed interventions help

A randomized invitation to a bundled COPD affordability and medication-management program involved 19,113 Medicare Advantage beneficiaries; low-income subsidy beneficiaries were excluded. The intention-to-treat effect on proportion of days covered was 3.8 percentage points (95% CI, 3.1-4.5). This refill-based measure is not inhaler use or clinical control. Estimates for COPD exacerbations and total spending were imprecise and did not demonstrate improvement. The bundled program does not isolate the contribution of information technology. [L-T01]

A Medicare Advantage real-time benefit-tool implementation study found no significant overall change in patient prescription spending; the difference-in-differences estimate was +1.2% (95% CI, -0.7 to 3.0). Monthly fill and total prescription spending estimates were also compatible with no change. In a separate, post hoc cluster-trial analysis, the Medicare-order fill difference was +0.5 percentage points (-2.4 to 3.4) overall. Its highest-cost drug-class subgroup showed +10.5 points (1.9-19.2), but the subgroup was exploratory and its Medicare denominator was not reported. The analysis shares a trial with an earlier cost-estimate report; these are not independent trials. [L-T02; L-T03; L-T03-P1]

No reviewed intervention establishes closure of a patient-sex disparity. These findings motivate testing a targeted, supported workflow against existing tools, with appropriate clinical outcomes and prespecified equity analyses. They do not justify assuming that price visibility alone creates benefit.

Translating the evidence into testable opportunities

The intervention evidence and current policy assessment are reported in the accompanying opportunity records and Table 2. Figure 6 maps the six unranked domains to proposed workflows and evaluation targets. Medication-affordability and post-fracture coordination have concrete intervention precedents within the selected literature. They warrant prospective testing of specified workflows, rather than claims that a new digital product has already been validated. Authorization, equipment fulfillment, kidney-care coordination and valuation research have different evidence gaps and should not inherit the effectiveness of an intervention in another domain.

A useful product hypothesis must name the clinical action, the impediment it addresses and the outcome that would improve. A benefit tool might change prescribing cost; a coordination service might improve appropriate testing; a research platform might reveal coding or matching errors. Faster processing, more covered claims or more billed services can be useful operational measures, but none alone establishes clinical benefit. Commercial assessment must separately verify who bears the problem, who can purchase a solution and whether value persists after implementation costs and existing alternatives are considered.

Current policy creates implementation opportunities without establishing product effectiveness. CMS-0057-F finalized non-drug prior-authorization requirements for specified payers, including Medicare Advantage, with operational provisions generally beginning in 2026 and API requirements generally beginning in 2027. The April 2026 CMS-0062-P drug expansion remains proposed as of this assessment. [P-046; P-048] Part D's 2026 \$2,100 out-of-pocket threshold applies to covered drugs; it is distinct from premiums, noncovered purchases and monthly cash flow. The Medicare Prescription Payment Plan spreads payments across the year without reducing total drug costs. [P-051; P-052] Likewise, coverage of clinical services or equipment does not establish separate payment for software; lymphedema furnishing-related education is bundled in supplier payment. [P-054] Product evaluation should therefore measure appropriate care and patient burden, while separately testing purchasing demand and implementation cost.

Discussion

Implications for researchers and developers

The strongest conclusion is methodological and practical: opportunities should be selected around a demonstrated mechanism of unmet clinical need. A broad allegation of gender bias is an inadequate product specification. It can obscure problems affecting men, confuse reproductive anatomy with shared disease, and encourage interventions that improve billing without improving care. Conversely, the absence of a matched male-female price difference does not rule out an inequity in diagnosis, referral, affordability or follow-through.

Within the selected evidence, medication-affordability workflows and post-fracture coordination are defensible candidates for pilot design because intervention precedents exist and measurable pathways can be specified. This is an inference about research readiness, not a ranking of expected health benefit, investment return or national market size. The intervention, data source, staffing, setting and patient population must resemble the proposed deployment closely enough to support transfer. A positive average result in Medicare Advantage may not generalize to Original Medicare or to beneficiaries excluded from the study.

The opportunity framework in Figure 7 requires evidence about need, mechanism, intervention, benefit and equity, and implementation. A project should define a patient-relevant primary outcome, a clinically appropriate denominator and potential harms before selecting a technology. Developers should test against usual care and credible existing services. Equity analysis should estimate differences in absolute benefit and reach across relevant populations, with interaction estimates where justified. Separate within-group significance tests do not establish a between-group difference. Digital access, language, disability and caregiver involvement may affect reach and should be measured rather than assumed.

Commercial validation is distinct work. A coverage code may pay for clinical labor or equipment without paying for software. An API requirement may enable interoperability without creating a purchaser or a viable margin. Proposed business models therefore need documented procurement interviews, implementation costing, data-access agreements and comparative value assessment. None has been completed in this project. Patient outcomes and equity should remain explicit success criteria even when an institution's initial purchasing rationale is administrative efficiency.

Limitations and completion requirements

The selected register and targeted intervention retrieval are not a systematically screened sample of all relevant evidence. One database's development searches, even with broad coverage and retained exports, cannot establish completeness. Keyword ordering can misclassify relevance; every tier is retained. Full-text access varies, cohort overlap remains incompletely adjudicated, and formal independent appraisal is unfinished. The present scope also leaves identity-unreported patient studies outside confirmed-cisgender primary estimates. That limitation cannot be resolved by relabeling recorded sex.

Heterogeneous outcomes, policy periods and comparison groups preclude a pooled estimate here. Historical use or spending may not describe current benefit design. Adjusting for income, service mix or utilization can remove part of an inequity pathway as well as reduce confounding; a defensible causal question must precede covariate selection. Selected intervention studies may improve process measures without demonstrating better health or reduced inequity. The opportunity domains are hypotheses derived after evidence inspection and have not undergone comparative commercial validation.

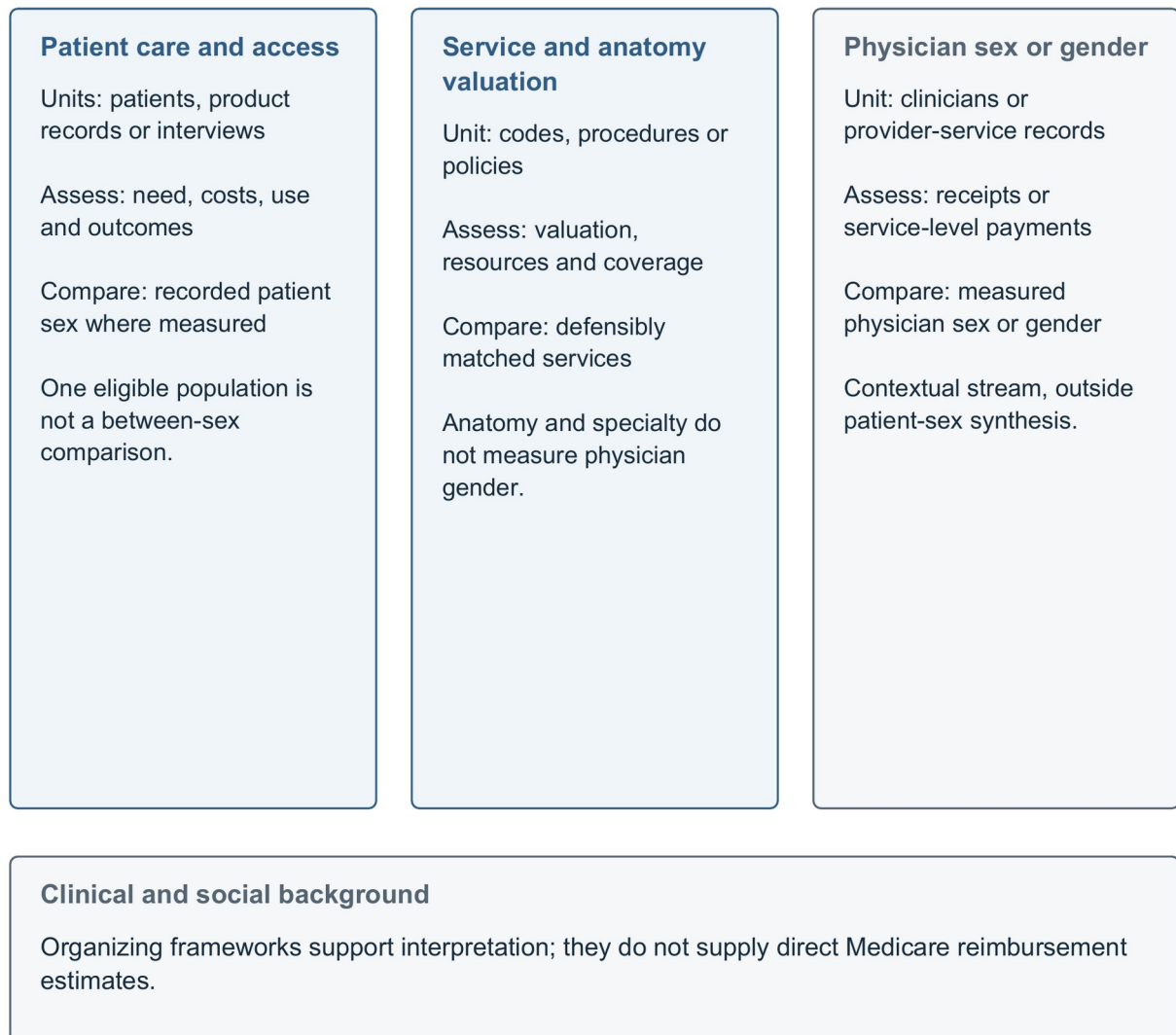
Before submission as a completed systematic review, the investigators must finalize the protocol amendments, complete independent search review and remaining source searches, adjudicate reports and eligibility, verify extraction, perform design-specific appraisal, and reconcile the selection flow. Outcome-specific certainty assessments should follow the actual evidence rather than a predetermined narrative. Authorship, funding, conflicts, contributions and final approval must be supplied by the responsible investigators. PRISMA reporting and polished presentation cannot substitute for these steps.

Conclusion

Medicare equity research can support useful health IT development when it connects clinically justified need to a specific financing or delivery mechanism and a testable intervention. The current evidence supports carefully designed studies of affordability, coordination and administrative processes, while preserving uncertainty and opposing findings. It does not yet establish a completed systematic review, a universal sex-based reimbursement penalty, or a ranked set of profitable ventures.

Figure 1. Separate the subjects of comparison

Analytical framework | Preserve units, measured variables and inference limits

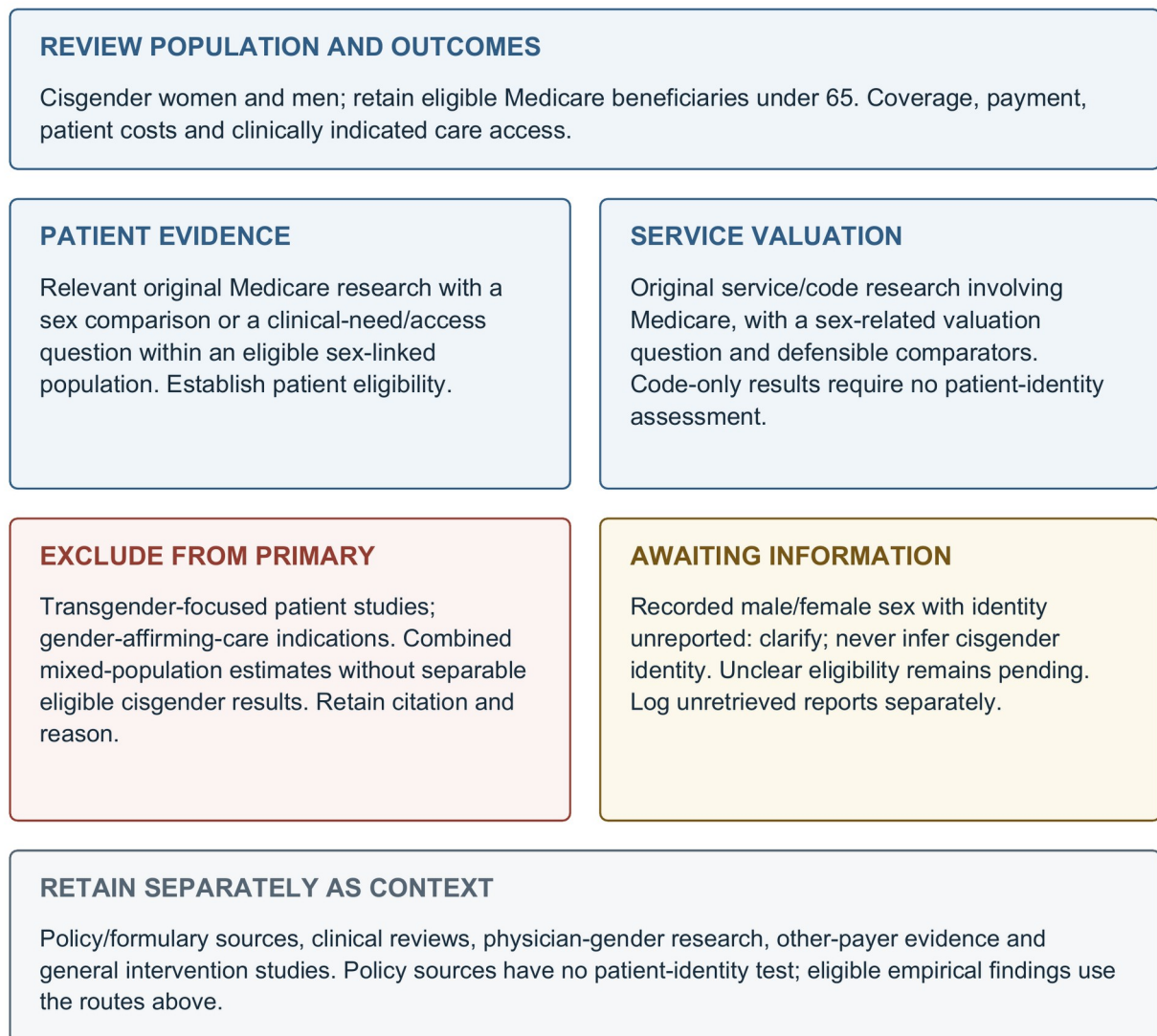


Preserve the measured variable, comparison axis and outcome. Do not infer patient disparities from clinician payments.

Three analytic streams answer different questions and require different units of analysis. Patient studies may compare recorded sex or describe care within one eligible population; neither recorded sex nor a diagnosis establishes gender identity. Service and anatomy studies compare procedures, resources or policies. Physician-gender studies compare clinicians and remain contextual to the patient-sex inquiry. Collections, salary and payment per service are distinct measures. Clinical reviews provide an organizing framework. Clinician interviews about patient barriers and prescriber datasets are classified by their analytic question, not by respondents' profession or data source. The diagram defines interpretation boundaries, not causal effects. [L-R01; L-P01; L-C03; L-Q01; L-B01; L-B01-E1]

Figure 2. Inclusion and exclusion criteria

Adopted scope: 16 Sep 2026 | Operational screening proposed: 21 Sep 2026

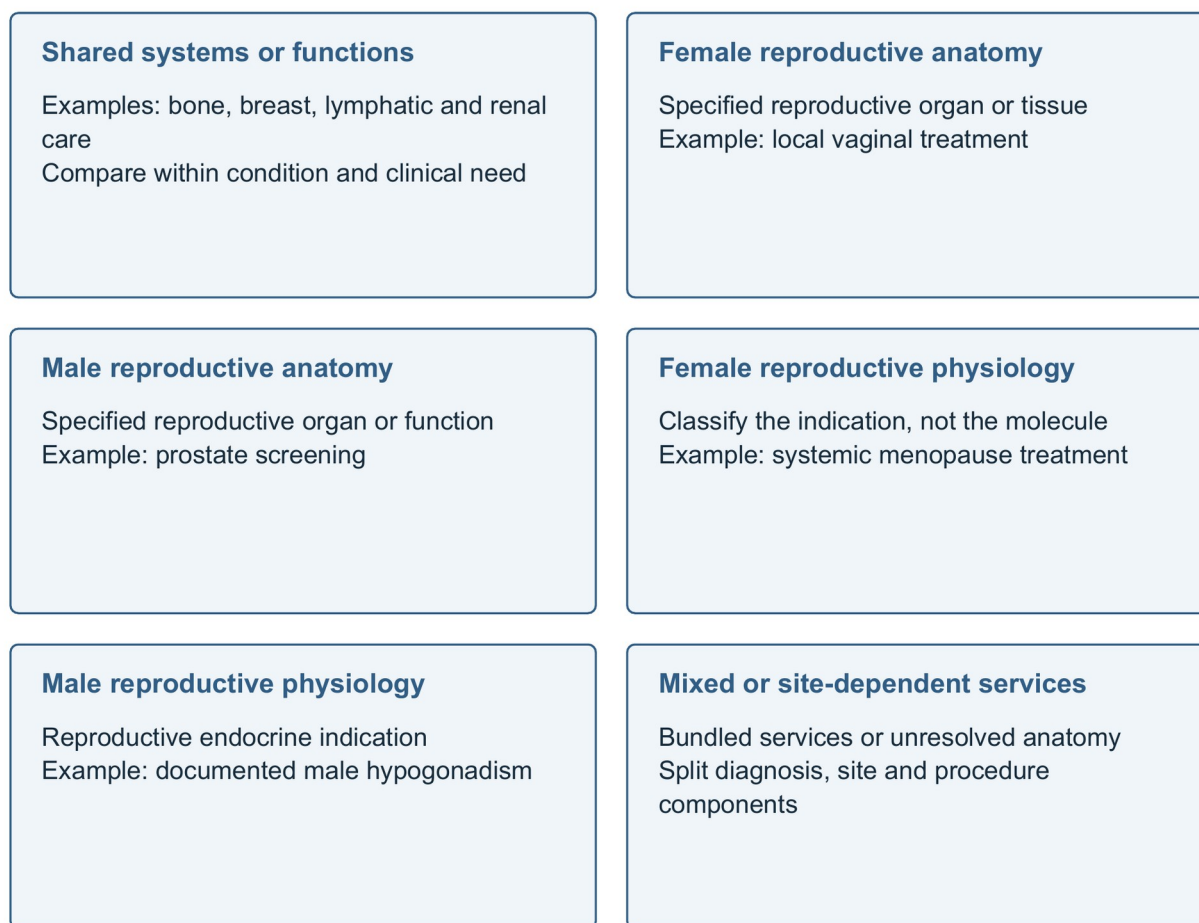


No opposite-sex counterpart is required. Retain null/opposing findings. No automatic exclusion by language or access. Final eligibility and selection counts remain pending.

The adopted September 16 scope is distinguished from operational screening proposed September 21. Original Medicare, Medicare Advantage and Part D results are extracted separately. Mixed populations contribute only separable eligible results; unreported gender identity remains unresolved rather than inferred. Code-only research has no individual patient-identity assessment, whereas any patient component requires one. Clinical need is assessed independently of coverage. Contextual records and exclusion reasons remain in the audit trail. The compact framework summarizes REVIEW_SCOPE.md and draft protocol sections 2, 3 and 6; it is not a completed PRISMA selection flow. [M-001]

Figure 3. Clinical anatomy and physiology

Classify each care indication | Categories describe services, not patient identity



Record separate descriptors

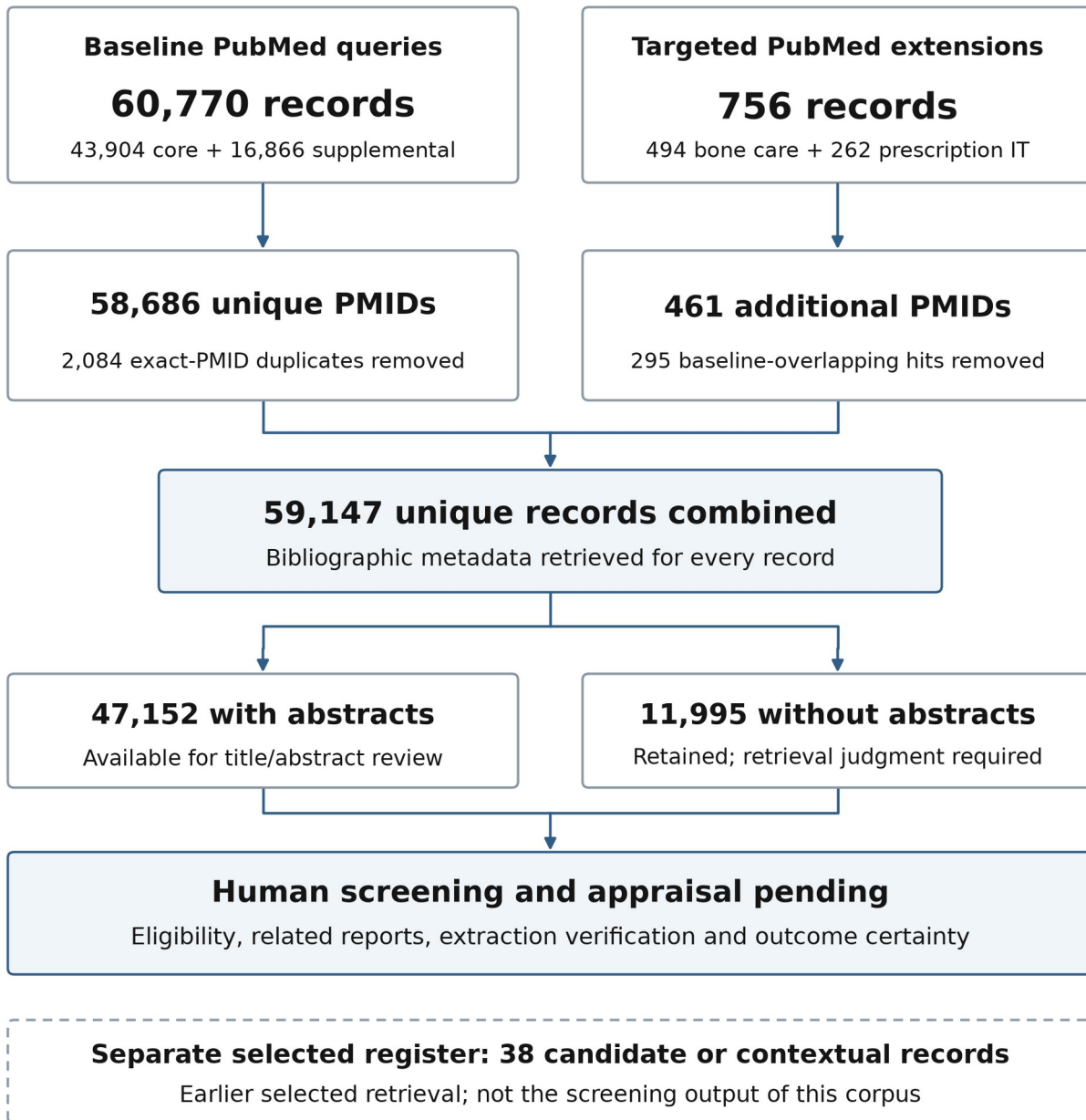
Policy eligibility: sex-defined rules. Disease burden: need and distribution.

These categories describe care, not gender identity. Define clinical need independently of reimbursement rules.

Care-item classification concerns the recorded indication and anatomical or physiological target. Shared-system care may have unequal disease burden or sex-defined policy eligibility without becoming exclusive to one sex. Reproductive anatomy and reproductive physiology remain distinct: local vaginal treatment and systemic menopausal treatment illustrate this distinction. Mixed or site-dependent services require component-level review before comparison. Policy eligibility and population burden are separate descriptors, not additional anatomical categories. Clinical need is defined independently of reimbursement eligibility, and an opposite-sex counterpart is not required for every indication. These categories neither assign patient identity nor quantify national inequity. [L-B01; L-B01-E1; C-001; C-002; C-003; C-004]

Figure 4. Retrieval and review status

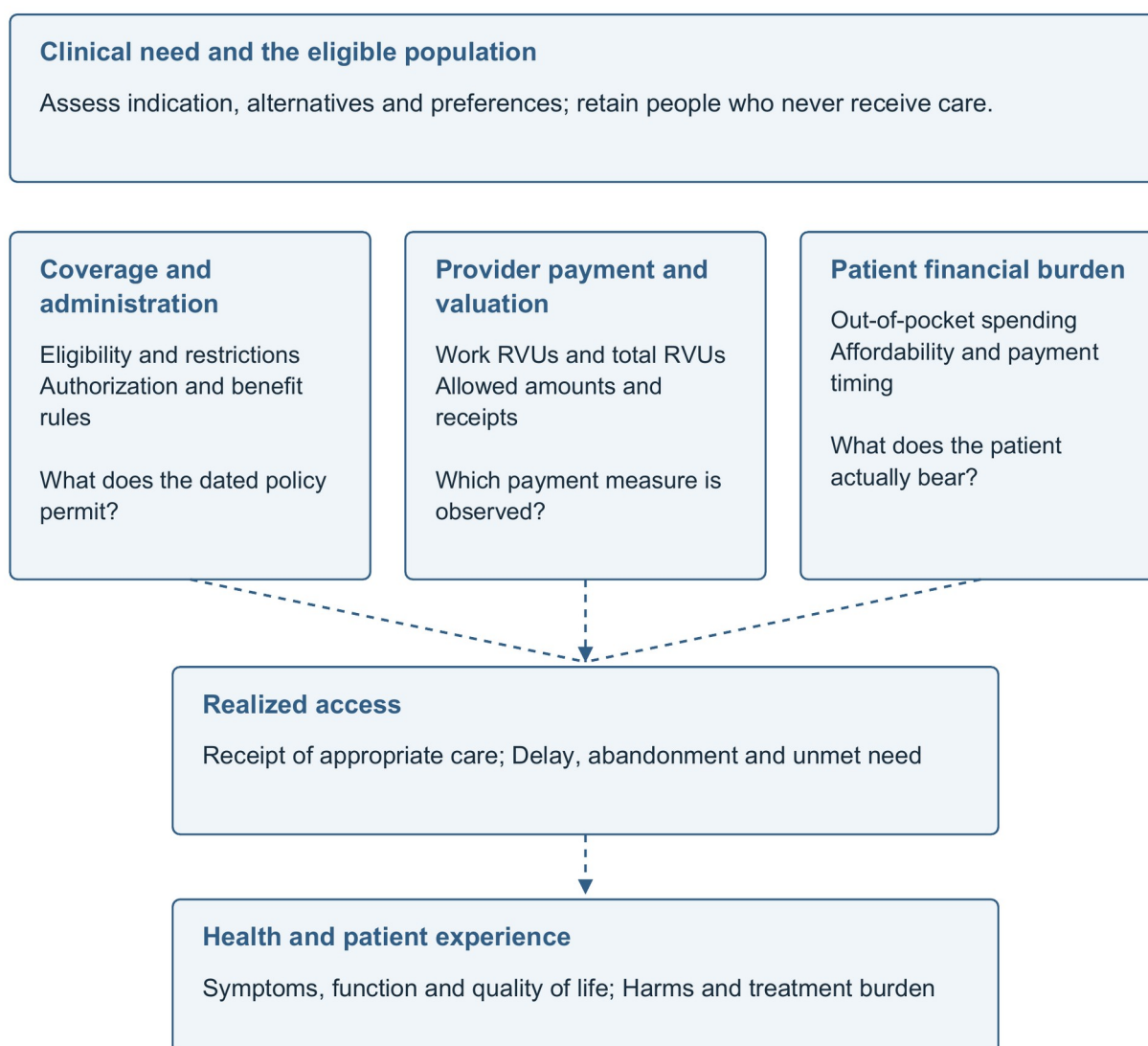
21 September 2026 | Development searches; selection review incomplete



Counts describe executed PubMed development searches as of September 21, 2026. Each unique PMID is a report record, not necessarily an independent study. Search extensions were developed after the initial search. No counts of full-text exclusions or final included studies are inferred. The selected 38-record register and targeted intervention retrieval are separate routes requiring later reconciliation. This is an audit diagram, not a completed PRISMA selection flow.

Figure 5. Financing and care outcomes

Measurement framework | Dashed links show relationships to evaluate



Links are hypotheses, with unestablished effects. Retain payer, policy year, age and entitlement. Assess equitable reach separately from average benefit.

Financing, access and patient outcomes are distinct measurement domains. The dashed links identify relationships to evaluate; they do not assert that coverage, higher payment or lower recorded spending causes better care. Policy permission differs from authorization performance, clinician receipts and patient out-of-pocket burden. Appropriate receipt, delays and unmet need require patient-level ascertainment, including people who do not complete a transaction. Health and experience outcomes require separate follow-up. Clinical need, indication, alternatives and preferences inform interpretation across the pathway. Analyses retain payer, policy period, age and entitlement, and evaluate equitable reach alongside average benefit. [L-D01; L-R01; L-P01; L-T01; L-T02]

Figure 6. Health IT research opportunities

Six unranked domains | Workflows and evaluation targets are proposals

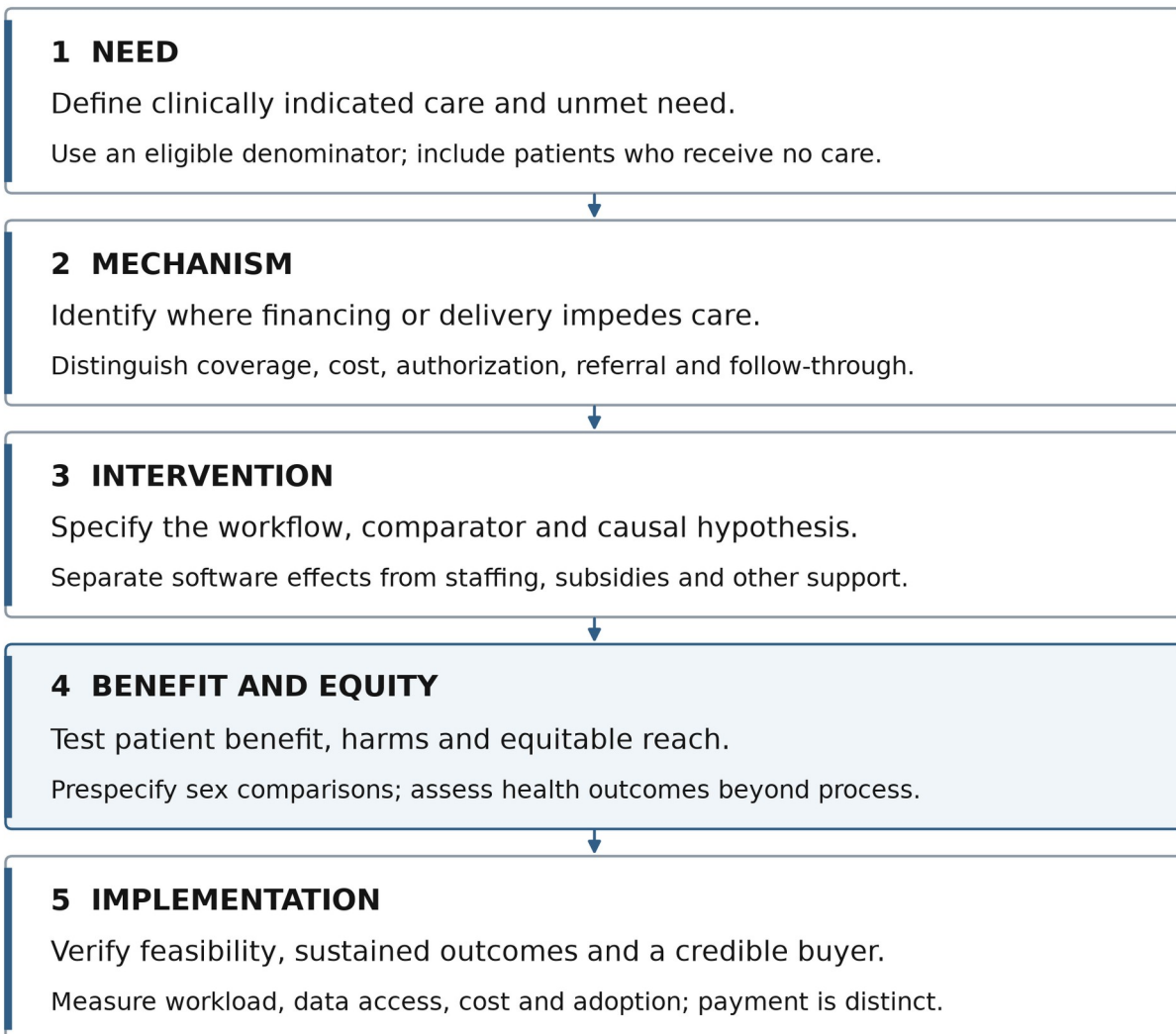
O-01 Medication affordability Test: staffed benefit navigation Measure: appropriate medication access and unmet need	O-02 Post-fracture follow-up Test: registry plus accountable human follow-up Measure: appropriate evaluation and shared decisions
O-03 Non-drug authorization Test: requirements and response tracking Measure: time to appropriate care and patient burden	O-04 Procedure-valuation research Test: transparent comparator and resource audit Measure: reproducibility and measurement reliability
O-05 Lymphedema item fulfillment Test: clinician-to-supplier coordination Measure: timely receipt, fit and ability to use	O-06 Kidney-care coordination Test: human-led education and practical support Measure: understanding and preference-concordant care

For every domain: compare existing services; measure harms, burden and equitable reach; verify implementation and buyer need.

Six purposively selected domains translate the local evidence register into prospective evaluation questions. The workflow and outcome in each panel are proposals, not established effects of a digital product. Medication and fracture-care interventions provide precedents for testing specified approaches; effectiveness does not automatically transfer to other domains. Procedure valuation initially requires reliable research measurements, with patient effects assessed separately. All proposals require comparison with existing services, clinically appropriate denominators, harms monitoring and explicit equity assessment. Coverage or an interoperability requirement does not establish software reimbursement, customer demand or profitability. Domain order conveys no priority or ranking. [L-T01; L-T02; L-G05; L-P01; L-P02; L-A02; L-Q01; P-054]

Figure 7. A health IT validation pathway

Proposed research framework | Evidence is required at each stage



Advance, revise or stop according to the evidence.

Average benefit does not establish equity; a coverage code does not establish a market.

Conceptual framework developed for this project after exploratory evidence review. Each stage requires a specified patient population, clinical indication and appropriate comparator. Progress may reveal that clinical support, policy change or an existing service is more useful than new software. Average benefit, equitable reach and commercial feasibility require separate evidence; no composite score or profitability ranking is implied.

Table 1 Selected evidence and limits of inference

Evidence and unit	Selected result	What the result cannot establish
L-R01 2016 Medicare survey Patient sex association	Female versus male adjusted odds of any cost-related nonadherence: under 65, 1.66 (95% CI 1.31–2.10); 65 or older, 1.21 (1.04–1.41).	Cross-sectional self-report; residual confounding and eligibility uncertainty. Not a current policy effect or causal bias estimate.
L-R02 Selected Chicago cohort Patient sex association	Adjusted persistent nonadherence OR 1.36 (0.95–1.95); transient nonadherence OR 0.81 (0.62–1.07), women versus men.	Both intervals include the null. Selected population; longitudinal outcomes differ from annual occurrence.
L-P01 and L-P02 Selected procedure pairs	L-P01: 55 pairs; male-associated work RVUs averaged 30% higher. L-P02: no significant aggregate work-RVU or RVU-per-minute difference.	Different matches and estimands. L-P02 abstract/companion pair counts conflict. Null is not equivalence; valuation is not patient access.
L-P03 Time-based procedure model	Female versus male procedure labels: median work RVUs/hour 10.6 versus 9.7; modeled specialty compensation \$555 versus \$599/hour.	Compensation multipliers are not Medicare conversion factors. Procedure/payer comparability and full-episode work remain concerns.
L-G05 Male fracture outreach 7,842 analyzed	BMD testing 10.7% versus 4.9%; adjusted OR 2.13 (1.74–2.62). Treatment initiation 4.0% versus 2.5%; adjusted OR 1.59 (1.19–2.12).	28.3% excluded after randomization. Bundled outreach; no fracture reduction, isolated software effect or sex-gap closure established.
L-T01 MA COPD invitation trial 19,113 randomized	Intention-to-treat proportion of days covered: +3.8 percentage points (3.1–4.5). Exacerbation and total-spending estimates did not establish benefit.	Refills are not inhaler use or clinical control. Low-income subsidy beneficiaries excluded. Intervention bundled cost sharing and management.
L-T02 MA benefit-tool implementation	Difference-in-differences patient prescription spending: +1.2% (–0.7 to 3.0). Total prescription spending and monthly fills also compatible with no change.	Nonrandom implementation; no demonstrated average improvement. No intervention-by-sex effect or equity benefit established.
L-T03 Post hoc cluster-trial report Medicare orders	Overall fill difference +0.5 percentage points (–2.4 to 3.4). Highest-cost drug-class subgroup +10.5 points (1.9–19.2).	Subgroup exploratory; its Medicare denominator unreported. L-T03-P1 is the same trial. Selected eligible orders; no equity-gap closure.

Examples from selected retrieval, not a complete included-study table or pooled analysis. Confidence intervals are 95%. MA, Medicare Advantage; OR, odds ratio; BMD, bone mineral density; RVU, relative value unit. Patient identity eligibility and formal independent appraisal remain unresolved. All numeric findings are provisional extractions with source locations retained in the working register and intervention evidence file.

Table 2 Unranked opportunities and the next useful study

Domain	Patient benefit or research contribution to test	Next validation and evidence boundary
O-01 Medication affordability	Resolve clinically appropriate prescriptions through benefit information plus pharmacist or clinician support; measure unmet medication need and disease-specific outcomes.	Pragmatic trial against existing tools and navigation. Bundled program benefit coexists with null average benefit-tool results. [L-T01; L-T02; L-T03]
O-02 Post-fracture follow-up	Complete appropriate bone assessment and treatment after a qualifying fracture, with accessible human follow-up.	Multisite trial; separate process improvement from later fracture outcomes. Existing male-only outreach has substantial post-assignment exclusions. [L-G05]
O-03 Non-drug authorization	Reduce avoidable delay and patient burden for clinically indicated services; preserve clinical review and alternatives.	Controlled rollout with all clinically eligible patients in the denominator. Final API rules enable workflows; they do not establish a sex disparity or product benefit. [P-046; P-047]
O-04 Procedure valuation research	Improve resource measurement, clinically defensible matching and reproducibility of code comparisons.	Independent full-episode work measurement and replication before policy-effect claims. Opposing findings require explanation. [L-P01; L-P02; L-P03]
O-05 Lymphedema fulfillment	Improve timely fitting, delivery and usable compression treatment; measure function and patient burden.	Post-2024 needs audit, then controlled fulfillment intervention. Historical costs precede the current Medicare benefit. [L-A01; L-A02; P-054]
O-06 Kidney education and support	Improve understanding, practical support and informed treatment choices at relevant kidney-disease stages.	US Medicare patient/caregiver needs assessment, followed by a controlled pilot. Existing clinician interviews are contextual, not measured Medicare outcomes. [L-Q01; P-055]

Domains are organized by clinical and implementation problem, not ranked by profitability, market size or expected health gain. Direct Medicare payment for the proposed software, purchasing demand, incremental cost-effectiveness and equitable reach are unverified. Detailed data requirements, prospective users/buyers and study designs are provided in Supplement B and the opportunity workbook.

Supplement A Methods and review completion

September 21, 2026. Supplement to the preliminary evidence synthesis, *Sex related differences in Medicare care and health IT opportunities*. This account separates work performed from methods proposed for the unfinished systematic review. It does not amend population eligibility or claim registration, completed human review, formal appraisal, or a completed systematic search across databases.

S1 Design scope and analytical roles

The adopted [review scope](#) addresses nationwide Medicare coverage, payment, patient costs, and access to clinically indicated care for cisgender women and men. Transgender-focused patient studies and gender-affirming indications are outside the current scope. Administrative male/female coding does not establish gender identity. Identity-unreported cohorts remain unresolved candidates and cannot support a confirmed-cisgender primary estimate. Investigator clarification about a separately labeled sex-recorded analysis is pending. Any change requires a dated amendment, a specified estimand, and consistent reconsideration of affected records; other work can continue under the existing criterion.

Patient comparisons, service/anatomy valuation, physician-gender context, and clinical/policy background are separate analytical streams. Shared systems/functions, reproductive anatomy, reproductive physiology, and mixed/site-dependent services retain their existing classification. Clinical need, disease burden, and sex-defined policy eligibility are distinct fields. Younger Medicare beneficiaries remain within scope; age, entitlement, payer component, indication, setting, and policy period are retained where available. Clinically eligible nonrecipients are relevant to access denominators, and lower observed use alone does not establish unmet need.

The active register contains 38 candidate or contextual records, not 38 systematically selected studies. A publication-role audit covers all 38 in [evidence_roles.json](#). Four additional intervention reports, L-T01, L-T02, L-T03 and L-T03-P1, are translational context outside that active register. The current scientific metadata set totals 53 records: 38 active, eight earlier scope exclusions, one linked Lancet correction, two outside-register leads/context records, and four intervention reports. L-T03 and L-T03-P1 describe the same trial, not two independent trials. The nine hypotheses and six opportunity domains were developed after preliminary evidence inspection and remain exploratory.

Eleven methodological references are retained separately in the canonical methodology register. The newly verified primary instrument references M-010 and M-011 do not increase the scientific-study count and do not indicate that appraisal has been completed.

S2 Identification and reproducibility performed work

PubMed through NCBI E-utilities was the sole bibliographic database platform searched in these development runs. MEDLINE is part of this PubMed source, not a second database. The September 21 corrected core yielded 43,904 records and the supplementary query 16,866; 2,084 exact-PMID overlaps gave 58,686 unique baseline records. Correcting the initial invalid formulary heading added 60 records without removing any. The original query/results remain preserved. The core did not require a sex/gender term; the supplementary strategy addressed sex/gender-related valuation and costs separately. A 6,769-record priority subset was exported first, followed by the remaining metadata; it was not an exclusion filter.

Post hoc bone-care and prescription/health-IT extensions yielded 494 and 262 hits. Their 756 distinct PMIDs overlapped baseline by 295, adding 461 and bringing the combined corpus to 59,147. All combined records have metadata: 47,152 with abstracts and 11,995 without. The baseline missed one of ten pilot sentinels selected after execution; the bone extension retrieved that source, PMID 18302495. This repair is a diagnostic observation, not

a validated recall estimate. No language, age, design, or full-text filter was applied. Date partitions were technical retrieval steps for ESearch's output limit, not substantive date exclusions.

Reproducibility component	Retained source
Exact baseline queries, translations, warnings and identifiers	Baseline queries, manifest
Baseline transport, recovery and archive verification	Execution audit, export script, normalization script
Exact extension queries and membership reconciliation	Extension queries, extension manifest, extension script
Current complete queue and rule-based ordering	Queue summary, queue preparation
Targeted intervention discovery and source access	Intervention search log, retrieval manifest
Original and current protocol	Pre-search snapshot, current protocol

The query JSON files are authoritative exact specifications; this supplement does not retype or simplify them. Raw successful responses, request parameters, timestamps, response hashes and source memberships are retained. The baseline audit reconciled 223 successful response bodies and 164 XML batches. Earlier request sidecars do not all name the HTTP method, and the archive is not a complete history of failed requests/retries. A reproducibility statement must preserve those logging limits.

Targeted web discovery, investigator-supplied papers and selected full-text retrieval remain separate identification routes. Eighteen title-pattern hits in the baseline were a purposive intervention queue, not full-corpus screening. Record membership and source routes require reconciliation before a final flow. The other protocol-listed databases, comprehensive citation searches and independent PRESS review remain unfinished. No absent abstract, queue tier, or keyword flag was used as a final exclusion.

S3 Record management and review independence

Exact-PMID deduplication is complete across the documented development queries. Seventeen shared-DOI groups and 303 normalized-title groups have been flagged, with no automatic merges; the groups may overlap and are not additional known duplicate counts. Reviewers must distinguish bibliographic duplicates, corrections, multiple reports, follow-up analyses and overlapping cohorts. Secondary trial reports retain their own report IDs and are linked to a study family. Companion results can inform different outcomes but must not contribute the same participants twice to a pooled comparison.

Fifty purposively sampled calibration records have been prepared. Human decisions remain blank; the independent human screening count is zero. Selected AI extraction, source-location checks and numerical validation have occurred, with access limits documented. They do not constitute two human reviewers or completed full-corpus selection.

The current protocol plans two human reviewers independently screening titles/abstracts and full texts, followed by reconciliation and third-person adjudication when needed. The calibration plan includes approximately 50 records and ten full texts. Potentially eligible reports not retrieved will be reported separately from assessed full texts excluded, with a defined primary-reason hierarchy. Existing pilot records are subject to the same final rules as newly identified records. Missing usable numerical data can prevent pooling without necessarily making an otherwise eligible study ineligible.

Critical eligibility judgments, outcome data, denominators and appraisal judgments will be independently extracted before reconciliation, retaining individual and consensus records. Bibliographic fields may be extracted once and verified by another reviewer as specified in the protocol. Full reports and relevant companion material, not abstracts alone, are needed for final methodological assessment. Sources with unresolved discrepancies remain flagged. AI assistance will be described by model/tool, task, date, procedure, human verification and limitations. These are the project's present commitments; this supplement does not imply that PRISMA universally requires exactly this staffing model.

Specific unresolved source issues are preserved in the [intervention extraction log](#). L-T03's main text and supplement differ on follow-up end dates, age-band labels, and excluded-order counts. Its intermediate-cost Medicare confidence interval appears inconsistently printed and was not silently corrected or used in the manuscript. The overall and highest-cost Medicare estimates were legible and separately extracted, but the highest-cost subgroup lacks a reported Medicare denominator, is exploratory, and comes from a small selected subset of trial orders. Its parent report is the same trial. These issues require reconciliation against the original protocol and, where needed, authorized author clarification before interpreting the subgroup as definitive. The L-T01 COPD report also contains a demographic-table arithmetic inconsistency, retained without inventing a corrected value. These are source discrepancies, not evidence of misconduct; uncertain fields remain explicit. No clarification request has been sent.

S4 Appraisal and synthesis planned methods

Formal risk-of-bias and certainty assessments have not been completed. Reviewers will select instruments by design and the result being interpreted. RoB 2 addresses randomized intervention effects; its parallel-group, cluster and crossover variants must not be treated interchangeably. An observational sex association from an RCT-derived cohort is not automatically a randomized contrast. The revised JBI analytical cross-sectional instrument addresses that design's exposure/outcome associations and does not substitute for cohort, prevalence, qualitative, quasi-experimental, or policy appraisal. Exact applicable versions, manuals, interpretation rules and pilot decisions must be documented before appraisal. The primary sources are Sterne et al. [M-010; PMID 31462531](#), and Barker et al. [M-011; PMID 40521701](#). Their bibliographic verification is not evidence that either tool has been applied.

Policy authority, effective dates, coding completeness, matching, setting and resources receive a separately described project appraisal. Work RVUs, total RVUs, allowed amounts, actual payments, provider receipts, total drug expenditure and patient out-of-pocket costs are different outcomes. Public formulary listings do not establish drug suitability, paid claims, affordability or unmet need. Original CMS analyses remain reproducible companion analyses, outside the literature's study-selection flow.

Structured synthesis by analytical stream, clinical indication, outcome, payer component and policy period is the default. The following **conditions for future pooling** operationalize the existing protocol; they do not retroactively make the nine current hypotheses prospective:

1. Establish a common question and estimand, including population, clinical need, exposure/intervention, comparator, outcome and time horizon, before selecting estimates by their results.
2. Confirm independent participant/study contributions or explicitly model dependence; reconcile report families and overlapping Medicare claims periods.
3. Establish compatible designs, measurement definitions and units. Keep randomized assignment effects, observational associations, modeled compensation and descriptive valuation separate. Distinguish medication possession, actual fills, clinical outcomes and patient costs. Do not combine odds ratios with risk ratios, or dollar spending with RVUs, without a justified transformation and common estimand.

4. Verify denominators, uncertainty and adjustment sets. Analyze unadjusted contrasts and appropriately adjusted effects according to their distinct causal interpretations; do not select the most adjusted or most significant result automatically. Treat possible mechanisms separately from confounders.
5. Obtain methods/statistical review of the planned model. For a compatible group, the existing protocol proposes random effects with REML and suitable small-sample inference; the precise method must be finalized before pooling. Report weights, uncertainty, heterogeneity and meaningful prediction intervals when estimable. Neither two studies nor a low heterogeneity statistic alone justifies pooling.
6. Prespecify sensitivity analyses for risk of bias, cohort overlap, policy period, payer directness and publication status. Subgroups require suitable data and direct interaction assessment where appropriate; separate within-group significance is not an interaction test. Post hoc subgroup findings remain exploratory.

Where pooling is inappropriate, present effect estimates, uncertainty, direction, clinical applicability and risk-of-bias limitations without vote counting statistically significant results. SWiM [M-007] is used where its intervention-effect scope fits, with any adaptation explicitly described. Lack of an asymmetry test does not establish absence of selective reporting.

GRADE [M-002] applies to suitable outcome/comparison bodies of evidence; it is not a numerical study score or a rating for every statute, inventory row or procedure code. The methodological approach must fit the effect/association question. GRADE-CERQual [M-008] concerns confidence in findings from an actual qualitative evidence synthesis, not a grading label assigned to a single interview paper. Descriptive maps and non-effect valuation summaries should state source confidence, comparability and measurement limits without mislabeling them formal GRADE ratings. No risk-of-bias, certainty or opportunity score is inferred from study size or statistical significance.

S5 Reporting extensions

PRISMA-S is already retained as **M-004**, Rethlefsen et al., [PMID 33499930](#); a duplicate citation is unnecessary. Final reporting will identify each database and platform, execution date, exact strategy/limits, supplementary route, peer review, updates, per-source counts, exports and deduplication procedure, with links to stable archived material. These reporting fields do not establish search completeness. PRISMA 2020 [M-001] and PRISMA-Equity [M-006] remain the existing reporting frameworks.

S6 Publication completion checklist

The unchecked items below are required work under the current protocol, not claims of completion. A preliminary report may accurately describe its limits; a completed systematic-review designation requires the completed review methods and evidence.

- Completed: Preserve development queries, source exports, corrections, count reconciliation and source-access logs.
- Completed: Separate active-register, metadata, report-family, context, hypothesis and original-analysis counts.
- Completed: Prepare source-located candidate extraction and a review queue while preserving all unscreened records.
- Pending: Finalize scope/estimands and date any authorized amendment; resolve individual eligibility consistently.
- Pending: Complete information-specialist development, independent PRESS review, planned source searches/citation routes, and a submission-period search update.
- Pending: Complete report-family/cohort reconciliation and independent human calibration, selection, full-text reasons and adjudication.

- Pending: Complete independent critical-data extraction, source-discrepancy resolution and design/result-specific appraisal.
- Pending: Finalize compatible synthesis groups and statistical methods before pooling; otherwise complete structured non-pooled synthesis.
- Pending: Complete appropriate certainty assessments and distinguish measured care gaps, causal evidence, intervention benefit and commercial assumptions.
- Pending: Reconcile the actual record/report/study flow; complete reporting checklists with final manuscript/supplement locations.
- Pending: Freeze a reproducible, permitted data/code release; finalize author contributions, AI use, funding, conflicts, ethics/data statements and author approvals.

The full [readiness audit](#) records evidence files and responsibilities. No registration, author contact, editor contact, stakeholder engagement, or submission is asserted. The protocol and translational assessment were developed after exploratory evidence work; later archiving cannot convert prior work into prospectively registered research.

Executed search sources

The complete original and corrected core queries, date partitions, returned translations, warnings, ID memberships and response hashes are retained under research/review_2026-09-21/pubmed. Targeted extension queries and raw responses are retained under research/publication_2026-09-21/search_extensions. The consolidated queue retains all 59,147 records without automatic exclusions. The following extension strings reproduce the submitted PubMed syntax.

PM-BONE-20260921-E1 Targeted bone-testing extension after diagnostic miss PMID18302495; post hoc development amendment

```
("Medicare"[Mesh] OR medicare[tiab]) AND ("Osteoporosis"[Mesh] OR "Bone Density"[Mesh] OR "Densitometry"[Mesh] OR osteoporos*[tiab] OR "bone mass measurement"[tiab] OR "bone mineral density"[tiab] OR DXA[tiab] OR DEXA[tiab])
```

PM-RXIT-20260921-E1 Targeted health-IT translation extension following new entrepreneurship purpose; not a comprehensive technology review

```
("Medicare"[Mesh] OR medicare[tiab]) AND ("real time benefit"[tiab] OR "real-time benefit"[tiab] OR "prescription benefit"[tiab] OR "electronic prescribing"[tiab] OR "e-prescribing"[tiab] OR ((drug*[tiab] OR medicat*[tiab] OR prescrib*[tiab]) AND ("decision support"[tiab] OR "price transparency"[tiab] OR "cost transparency"[tiab] OR "cost information"[tiab])))
```

Supplement B Opportunity assessment details

These six domains are research hypotheses without a comparative investment ranking. The accompanying workbook preserves the full evidence, workflow, source and policy fields. Prospective users and buyers have not been interviewed or validated. Proposed trials are future studies, not registrations or commitments. The patient population and separate measurement of sex and gender identity require an explicit protocol.

Opportunity 1 Prescription affordability and benefit navigation

O-01 Patients may be unable to obtain a clinician-selected medication because its actual out-of-pocket cost, coverage requirements or feasible alternatives are unclear at prescribing. Displaying a price alone may not resolve that barrier.

Clinical basis: GSM involving vaginal anatomy and menopausal physiology remain distinct from shared respiratory-system COPD. Hormone prescriptions are not interchangeable by name or price.

Intervention to test: Clinician/pharmacist-supported navigation, with an embedded benefit-information module.

Patient benefit: Primary: clinically appropriate prescription initiation/continuation and patient-reported unmet medication need.

Evidence anchors: L-D01 (Historical Medicare descriptive signal); L-D03 (Clinical counterevidence); L-D04 (Clinical background review); L-T01 (Randomized invitation to a bundled affordability/adherence program); L-T02 (Null implementation evidence); L-T03 (Exploratory post hoc cluster-trial analysis).

Prospective users: Patients and caregivers with permission; Pharmacists; Prescribers and medication-access staff. **Prospective buyers, unverified:** Health systems; Part D/MA plan care-management teams; Pharmacy organizations.

Data requirement: Current enrollment, formulary, strength/route/NDC or RxNorm, pharmacy network and patient-specific benefit data.

Policy boundary: NCPDP RTPB v13 requirements and existing beneficiary tools create technical context. The 2026 \$2,100 covered-Part-D threshold does not cap all costs; the Prescription Payment Plan changes timing, not price. [P-049; P-050; P-051; P-052; P-056; P-057]

Validation design: Prospectively registered pragmatic cluster trial comparing existing RTBT/usual navigation with RTBT plus staffed closed-loop navigation; specify any cost-sharing assistance as a separate component.

Analysis: Primary intention-to-treat analysis, with uptake measured separately. Prespecify clinically high-cost strata without relying on a post hoc subgroup; report attrition and missing cash purchases. Stratify recorded sex, indication, age, disability entitlement, dual/LIS status and digital access without claiming identity from recorded sex.

Success criterion: Improved clinically appropriate access and patient experience without worse clinical control or avoidable adverse events; monetary savings alone are insufficient.

Important risk: Existing plan tools, pharmacy counseling and human navigation may achieve the same result at lower cost.

Unresolved evidence: No verified evidence that this proposed product closes a national patient-sex access gap. Incremental effectiveness after Part D redesign and cost-effectiveness remain unknown.

Opportunity 2 Post fracture bone health follow up

O-02 Some patients with a fragility fracture do not receive timely evaluation or an appropriate secondary-prevention plan. A referral can be lost between acute care, primary care and specialty care.

Clinical basis: Shared skeletal system; sex-specific risk and eligibility categories are separate from anatomy.

Intervention to test: Fracture-event registry plus human follow-up.

Patient benefit: Primary: completed clinically appropriate post-fracture evaluation and documented shared treatment decision within a prespecified interval.

Evidence anchors: L-G05 (Direct randomized outreach evidence in men); L-G04 (Historical utilization context).

Prospective users: Fracture-liaison nurses; Orthopedic and primary-care teams; Patients/caregivers. **Prospective buyers, unverified:** Hospitals and integrated systems; MA plans; Accountable care organizations.

Data requirement: Admission/discharge notifications, fracture site/trauma confirmation, prior DXA and treatment history.

Policy boundary: Part B bone measurement uses specified criteria and frequency rules. USPSTF's insufficient evidence for screening men concerns adults without known osteoporosis or fragility-fracture history; it must not be used to block post-fracture evaluation. [P-012; P-053; P-056; P-057]

Validation design: Multisite pragmatic cluster trial of closed-loop follow-up versus existing discharge/referral processes, enrolling both sexes with clinically comparable qualifying fractures.

Analysis: Preserve all randomized patients in the principal analysis; report deaths, loss to follow-up and competing risks. Prespecify age/entitlement strata, baseline risk and sex interaction; do not power the trial only for testing uptake.

Success criterion: Meaningful access/process improvement without unnecessary testing, followed by adequately powered assessment of fractures and function.

Important risk: A staffed fracture-liaison service without new software may be sufficient.

Unresolved evidence: Digital incremental benefit, contemporary multi-payer effectiveness, fracture reduction and sex-specific benefit remain unestablished.

Opportunity 3 Prior authorization workflow for clinically indicated non drug care

O-03 Incomplete documentation, unclear requirements and untracked responses can interrupt a clinically indicated service. Current interoperability requirements create a testable workflow opportunity, but do not prove that a particular tool improves care.

Clinical basis: Service-specific/shared or mixed anatomy; do not label all authorization requests as female-specific or infer physician gender.

Intervention to test: Traceable requirements discovery, documentation preparation and response tracking.

Patient benefit: Primary: time from clinical decision to appropriate care, including abandoned/unsubmitted requests.

Evidence anchors: L-A02 (Policy-level authorization context); L-Q01 (Qualitative access-mechanism context only).

Prospective users: Clinical authorization staff; Ordering clinicians; Patients/caregivers tracking status.

Prospective buyers, unverified: Health systems and practices; Payers; Existing EHR or transaction-service organizations.

Data requirement: Payer endpoint/authentication and applicable API capability; coverage requirement provenance and version.

Policy boundary: CMS-0057-F generally has 2026 operational and 2027 API dates for impacted payers, including MA, and excludes drugs. The 2026 CMS-0062-P drug expansion remains a proposal in this assessment. Original Medicare pathways and Part D pharmacy NCPDP workflows are separate. [P-046; P-047; P-048; P-050; P-059; P-056; P-057]

Validation design: Prospective controlled rollout or cluster trial across defined services, with parallel clinically eligible patient ascertainment and audit of requests requiring manual handling.

Analysis: Prespecify clinically meaningful delay thresholds, urgency, eligible population and policy versions. Compare rates by recorded patient sex after appropriate clinical-need stratification; do not infer causation from raw denial rates.

Success criterion: Reduced clinically important delay and burden with no increase in incorrect submissions, inappropriate treatment or exclusion of difficult cases.

Important risk: Existing payer/EHR/clearinghouse integration or better staffing may outperform a new product.

Unresolved evidence: No extracted study establishes this proposed product's patient benefit, cost-effectiveness or sex-gap closure. National administrative metrics do not supply a patient-level causal test.

Opportunity 4 Procedure valuation research and audit infrastructure

O-04 Studies reach different conclusions when they compare broad anatomical analogues, tightly matched procedures, operative-time-normalized RVUs or specialty compensation. The resource basis of selected payment comparisons remains uncertain.

Clinical basis: Anatomy, shared functions, procedure approach and clinical purpose must be classified separately. Specialty is not physician sex and an RVU is not patient access.

Intervention to test: Reproducible research dataset and transparent comparator audit.

Patient benefit: Near-term research outcomes: reproducibility, comparator agreement, completeness of full-episode resource measures and uncertainty.

Evidence anchors: L-P01 (Broader anatomical comparison); L-P02 (Null evidence with closer valuation matching); L-P03 (Metric-dependent counterevidence); L-P04 (Resource-measurement context).

Prospective users: Health-services researchers; Clinical coding/resource experts; Patient advisory panels.

Prospective buyers, unverified: Research funders and academic consortia; Specialty organizations; Payers or policy evaluators.

Data requirement: Effective-date CMS files, CPT licensing/definitions, setting, modifiers, geographic indices and global periods.

Policy boundary: CMS RVU files are inputs for research, not coverage determinations. The July 2026 landing page identifies separate QP/non-QP files; every payment model requires source-version and eligibility checks. [P-035]

Validation design: Registered multicenter measurement study with independently adjudicated comparators, full-episode work sampling and external replication; only subsequently evaluate downstream policy changes using a justified causal design.

Analysis: Separate code-level from patient-level estimands; report null pairs, unmatched indications, alternative matching rules and uncertainty. Avoid counting related historical analyses as independent replications.

Success criterion: Reliable measurements and conclusions robust to reasonable comparator choices; a direction-neutral result is valid.

Important risk: An open academic dataset and analysis scripts may be preferable to commercial software.

Unresolved evidence: The three headline contrasts are not directly poolable; contemporary full-resource measurements and causal downstream access effects are missing.

Opportunity 5 Lymphedema compression item access and fulfillment

O-05 Coverage may not translate into timely receipt of a properly fitted, usable compression item. Orders, supplier availability, measurements, cost sharing and replacement needs may require coordination.

Clinical basis: Shared lymphatic system. Breast-cancer survivorship is one clinical context, not a sex-exclusive anatomy or coverage category.

Intervention to test: Clinician-to-supplier fulfillment and patient-use follow-up.

Patient benefit: Primary: time to receipt of the prescribed, correctly fitted item and ability to use it.

Evidence anchors: L-A01 (Historical patient-cost signal); L-A02 (Historical coverage context).

Prospective users: Patients/caregivers; Lymphedema therapists; Ordering clinicians; DMEPOS supplier staff.

Prospective buyers, unverified: Cancer-survivorship programs; Integrated care systems; Enrolled suppliers or payer care-management programs.

Data requirement: Diagnosis/order, item and body-part details, clinician-reviewed fitting needs and supplier enrollment/network data.

Policy boundary: The current Part B benefit covers qualifying garments/wraps/bandaging, with general deductible/coinsurance requirements. Supplier furnishing-related education is bundled; coverage does not create separate payment for this software. [P-021; P-054; P-056; P-057]

Validation design: Prospective post-2024 access audit followed by a controlled or randomized fulfillment intervention across diverse suppliers and clinical sites.

Analysis: Include people with orders who never receive items; separate primary from secondary lymphedema, affected site, clinical severity and plan type. Evaluate sex differences within comparable need; do not extrapolate from breast-cancer-only samples.

Success criterion: Improved appropriate item receipt/use and function without higher inappropriate ordering, patient costs or skin injury.

Important risk: Therapist/supplier case management may resolve most failures without new software.

Unresolved evidence: Current national fulfillment, fit and cost-sharing barriers after the benefit expansion have not been established by the cited studies.

Opportunity 6 Kidney care education and practical support coordination

O-06 Caregiving demands, transport, finances and information needs may impede access to kidney care. The available qualitative evidence identifies hypotheses; it does not measure current Medicare barrier prevalence.

Clinical basis: Shared renal system. Patient sex/gender may affect barriers, but interviewed clinicians' gender is not the analytic exposure.

Intervention to test: Human-led education/referral coordination with a patient-facing task and preference record.

Patient benefit: Primary: patient understanding, preference-concordant decisions and completion of clinically appropriate education.

Evidence anchors: L-Q01 (Qualitative mechanism evidence).

Prospective users: Patients/caregivers; Nephrology nurses and educators; Social workers; Care coordinators.

Prospective buyers, unverified: Nephrology organizations; Integrated kidney-care programs; Payer care-management teams.

Data requirement: Clinician-confirmed stage/indication and referral, education attendance and patient-reported goals.

Policy boundary: Part B kidney-disease education covers up to six eligible stage-4 sessions with referral/provider conditions and cost sharing. This is not universal coverage for navigation, transport or a standalone application. [P-055; P-056; P-057]

Validation design: Begin with US Medicare patient/caregiver interviews and an access audit, then co-design and pilot a controlled coordination intervention before a definitive pragmatic trial.

Analysis: Include people declining services and report negative/contradictory accounts. Prespecify sex, age/entitlement, rural access and socioeconomic strata; do not infer identity from renal diagnosis or sex fields.

Success criterion: Improved understanding and preference-concordant care with reduced burden; modality uptake or billing volume alone is not success.

Important risk: Additional social-work/education capacity may matter more than software.

Unresolved evidence: No directly evaluated Medicare technology intervention or verified patient-sex benefit in the cited record. Patient and caregiver evidence is required before product selection.

Supplement C Reference roles and reporting declarations

The legacy register contains 31 broadly labeled candidate records and seven contextual records. These labels are not final eligibility decisions. A separate publication-role audit identifies the following provisional uses; directness does not resolve patient identity or methodological validity.

Potential Medicare patient empirical evidence: 9 records. L-G04, L-G05, L-G09, L-G10, L-H01, L-R01, L-R02, L-R03, L-R04.

Medicare descriptive drug use and spending: 1 records. L-D01.

Potential Medicare service valuation evidence: 6 records. L-P01, L-P02, L-P04, L-P05, L-P07, L-P08.

Service valuation inputs with unresolved Medicare directness: 3 records. L-P03, L-P06, L-P09.

Clinical efficacy or safety background: 10 records. L-P10, L-D02, L-D03, L-D04, L-D05, L-D06, L-D07, L-D08, L-D09, L-D10.

Other or mixed payer background: 2 records. L-A01, L-A02.

Physician gender context: 3 records. L-C01, L-C02, L-C03.

General or qualitative context: 4 records. L-B01, L-B02, L-B03, L-Q01.

Three newly examined intervention study families are retained separately as translational context. L-T03 and L-T03-P1 are reports of the same trial. The bibliography preserves excluded records and corrections; a correction is not an independent clinical study.

Authorship funding and interests

Author names, affiliations, corresponding author, contributions, funding and competing interests have not been supplied or approved. No absence of conflicts or funding is asserted. Responsible investigators must review the complete manuscript, verify the scientific judgments and approve any submission. No journal inquiry or submission was sent in preparing this package.

Use of artificial intelligence

OpenAI Codex assisted with search development and execution, public-source retrieval, preliminary screening notes and data extraction, bibliographic checking, code, evidence organization, prose and figures. Parallel AI checks were used for arithmetic, citation consistency and document review. These checks do not constitute two independent human reviewers or completed formal appraisal. Investigators retain responsibility for verification, eligibility decisions, interpretation and the final manuscript. The disclosure should be updated to reflect actual human contributions and the submission journal's requirements.

Data code and access

The repository retains queries, successful raw response captures, identifiers, hashes, provisional extractions, source locations, exclusions, bibliography, RIS export and artifact builders. Public-source availability varies; abstract-only and targeted full-text access are identified in the register. Third-party publication and data rights remain with their owners. There are no private patient-level records in this package. Any institutional ethics determination required for future patient or developer research remains to be obtained.

Repository: [clinical-trials/medicare](#)

References

Stable IDs support the working manuscript until final journal numbering. Numeric PMIDs identify scientific and methodological publications; policy/data sources have no PMID. The separate living bibliography and RIS retain all active, excluded, contextual and correction records. Bibliographic verification is distinct from eligibility and full-text review.

[L-B01] Mauvais-Jarvis F, Bairey Merz N, Barnes PJ, Brinton RD, Carrero JJ, DeMeo DL, et al. Sex and gender: modifiers of health, disease, and medicine. *Lancet*. 2020;396(10250):565-582. PMID 32828189. DOI 10.1016/s0140-6736(20)31561-0. [Source](#)

[L-B01-E1] Department of Error. *Lancet*. 2020;396(10252):668. PMID 32891210. DOI 10.1016/s0140-6736(20)31827-4. Correction linked to L-B01; not an independent study. [Source](#)

[P-041] Centers for Medicare & Medicaid Services. U.S. Personal Health Care Spending By Age and Sex: 2020 Highlights. Access/status: 2026-09-16. [Source](#)

[P-042] Centers for Medicare & Medicaid Services. NHE Fact Sheet. Access/status: 2026-09-16. [Source](#)

[P-043] Centers for Medicare & Medicaid Services. Get started with Medicare. Access/status: 2026-09-16. [Source](#)

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